Week 2 · Laboratory Problems — close the speed loop in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 2 lecture hour
- Three problems, in order. Each one adds one block to the previous answer.

What this hour is for

Week 1 drove the hull with a command someone typed. This hour closes a loop around it, and the whole point is to see **what closing a loop can and cannot buy**.

The hull and the thrust map are provided. The thrust map is provided because turning a demanded force into two shaft speeds is Appendix A1's subject, and repeating it here would spend the hour on the wrong thing. Everything between the reference and the plant — the summing junction, the gains, the integrator — is built by hand.

Each problem ends with a **number measured in the lecture** and a **picture of the correct result**. There is no single correct diagram; the checker tests the physics.

Before starting

```
cd lectures/W02_simulink/problems
W02_P1_start                % creates W02_P1.slx — hull and thrust map only
W02_check(1)                % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

> **★ Two requirements on every model**
>
> - a To Workspace block named `xlog`, format `Structure With Time`, fed by the plant's twelve-state output
> - a second one named `Xlog`, same format, carrying the demanded surge force that enters `X to n`
>
> Two logs and not one, because Week 2's subject is the relation between a demanded **force** and the **speed** it buys. A log of the states alone cannot show that a proportional controller has run out of force.

Problem 1 · The open loop (20 minutes)

Build. A constant force straight into the thrust map, and two logs coming out.

```
Constant X_open  →  X to n  →  Otter USV  →  To Workspace (xlog)
```

Predict before running. In steady state the thrust balances linear surge damping, so

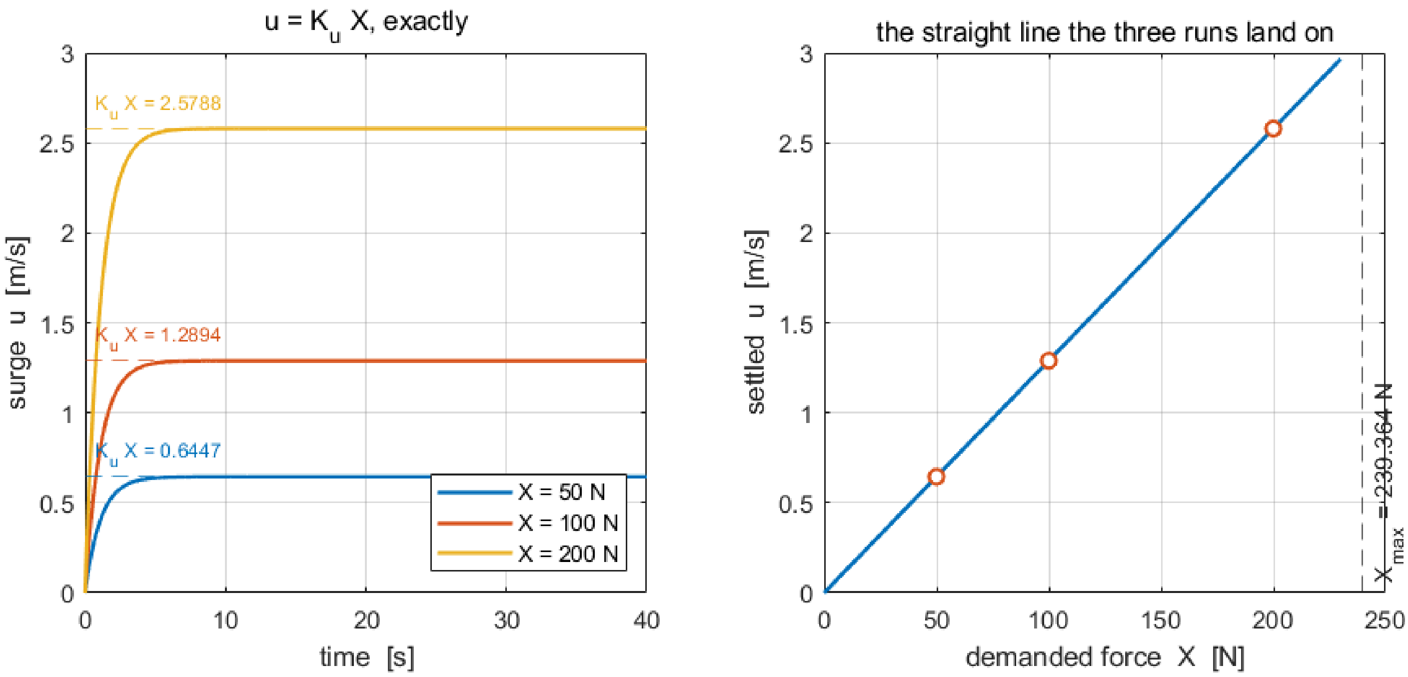
$$u_{ss} = K_u X, \qquad K_u = \frac{1}{|X_u|} = 0.012894\ (\mathrm{m/s})/\mathrm{N}$$

Write down what $X = 50$, 100 and 200 N should give.

Verify. `W02_check(1)`.

What the checker expects	
u_{ss} at $X = 50$ N	0.6447 m/s
u_{ss} at $X = 100$ N	1.2894 m/s
u_{ss} at $X = 200$ N	2.5788 m/s

What a correct model produces



Reading the figure	
left	three first-order rises, each landing exactly on its own dashed $K_u X$ line. No overshoot: there is no loop yet
right	the same three results as points on the straight line $u = K_u X$, with the actuator ceiling marked
the check	if the rises overshoot, something is already fed back; if they land off the dashed lines, the thrust map is being bypassed

The point. The relation is **exact**, not approximate. Surge damping in `otter.m` is linear, so the steady state is a straight line and not a curve that merely looks like one.

Problem 2 · Proportional control (20 minutes)

Build. Add the reference, the summing junction, and one gain.

```
Step u_d  →  (+)(-)  →  Kp  →  X to n  →  Otter USV
                ↑                               ↓
                └──────────  u  ←───────────────┘
```

The feedback signal is u**, state 1** — not the speed over ground $\sqrt{u^2 + v^2}$. With no current and no steering the two agree here, and in Week 4 they do not. A loop written against the wrong signal keeps working until exactly the moment it matters.

Predict before running. The final value theorem gives

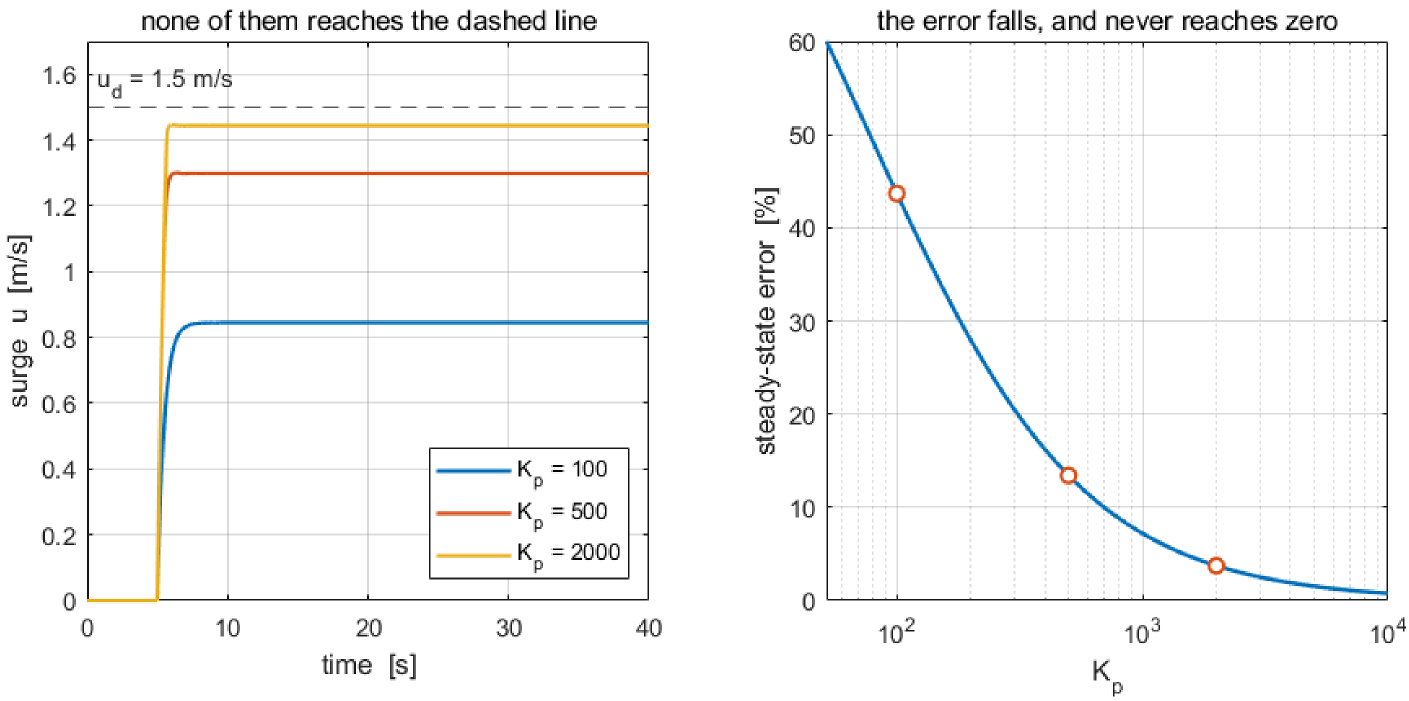
$$\frac{u_{ss}}{u_d} = \frac{K_p K_u}{1 + K_p K_u}$$

Work out the steady error at $K_p = 100$ before touching the model.

Verify. `W02_check(2)`, with $u_d = 1.5$ m/s.

K_p	expected u_{ss}	error
100	0.8448 m/s	44 %
500	1.2986 m/s	13 %
2000	1.4440 m/s	3.7 %

What a correct model produces



Reading the figure	
left	three settled values, and **none of them touches the dashed reference**
right	the measured points land on the theoretical curve $K_pK_u/(1 + K_pK_u)$, which approaches zero error and never arrives
the check	if any trace reaches 1.5 m/s, an integrator has been added early

The point. The plant has no free integrator, so the loop is **type 0**. The steady force that the damping demands can only be produced by a **non-zero error**. This is not a tuning failure and no value of K_p removes it.

Problem 3 · Add the integrator (20 minutes)

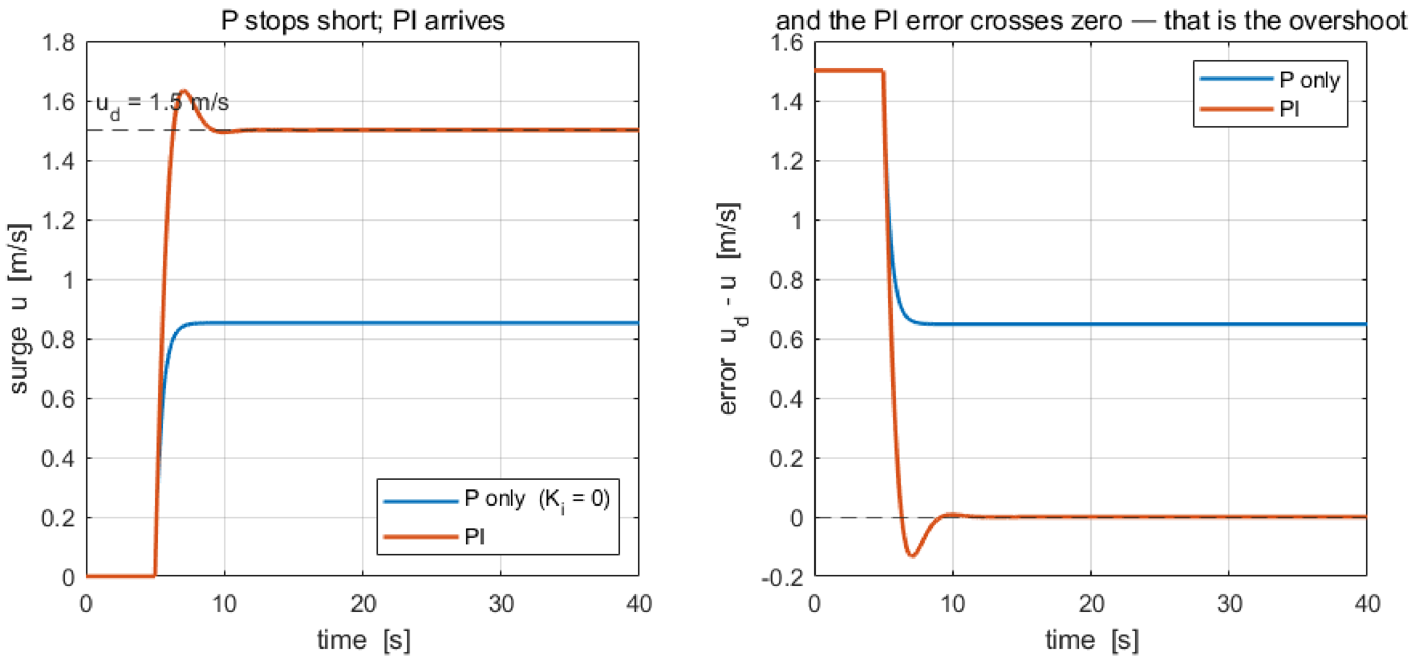
Build. One more branch: K_i into a **discrete-time** integrator, summed with the proportional term.

The integrator must be discrete because the model is fixed-step. A continuous integrator inside a fixed-step loop invites a solver-order mismatch that shows up as a slow drift rather than as an error message.

Verify. `W02_check(3)`, with $K_p = 102$, $K_i = 192.38$.

What the checker expects	
steady speed	1.5000 m/s
steady error	0, to tolerance
overshoot	**present** — the checker fails a response with none

What a correct model produces



Reading the figure	
left	the P trace stops short; the PI trace arrives at the dashed reference
right	the same runs as error. The P error settles on a non-zero value; **the PI error crosses zero and comes back**
the check	that crossing *is* the overshoot. A PI response with no crossing means K_i is not actually in the loop

The point. The integrator supplies the steady force that the damping demands, so the error no longer has to. The price is a state that keeps acting after the error has passed through zero — which is overshoot, and, when the actuator saturates, **windup**. Sections F to H of the lecture are about paying that price down.

Marking

	Weight	What is being marked
Problem 1	30	`W02_check(1)` passes, and the three predicted speeds were written down **before** running
Problem 2	40	`W02_check(2)` passes; the type-0 argument is stated in one sentence
Problem 3	30	`W02_check(3)` passes; the answer to "what did the integrator cost" is stated

A model that fails a check but whose written reasoning is right earns more than one that passes with no reasoning.

If something goes wrong

Symptom	Cause	Fix
`The model has no To Workspace block whose variable name is xlog`	the variable name is still `simout`	rename it
`xlog has 1 column`	a matrix signal reached the log	insert a Reshape set to `1-D array`
The speed settles at the reference in Problem 2	an integrator is present	set $K_i = 0$; Problem 2 is proportional only
The response drifts slowly upward with PI	a **continuous** integrator in a fixed-step model	use Discrete-Time Integrator with sample time `h`
Dimension error at the thrust map	the loop is feeding the whole 12-state vector back	select state 1 first
Every number is slightly off	the solver was changed	fixed-step `ode4`, `h = 0.02` s

Reference answers are in `../solutions/`. Read them **after** attempting the problem.